JENNIFER HSIA

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Machine Learning Ph.D. candidate working on LLM evaluation, model behavior, and reliability. Builds controlled evaluations that reveal when models reach conclusions their inputs don't justify, such as deferring to a user's unsupported claim or picking a plausible but wrong answer, and develops methods that reduce these failures.

EDUCATION

Carnegie Mellon University 2021–Present
Ph.D. in Machine Learning (expected May 2027) · M.S. in Computer Science (2023)
Advisors: Zachary C. Lipton and Aarti Singh

Princeton University 2017–2021
B.S.E. in Computer Science, magna cum laude · Minors in Statistics and Machine Learning

RESEARCH EXPERIENCE

Carnegie Mellon University | Graduate Researcher, Machine Learning Department 2021–Present
First or co-first author on all projects below.

Deference to user claims in LLM advice In submission

- Studying how users' claims about their own circumstances shape LLM advice.

Robust answer selection when retrieval favors a wrong answer EMNLP 2026

- Showed that retrieved evidence can make a wrong answer look compelling even when it misses a key constraint in the question, and, in controlled experiments, that adding passages supporting such answers makes LLMs more likely to choose them.
- Developed and implemented Joint Constraint Aggregation (JCA), which extracts shared constraints from a question and jointly evaluates candidate answers against them, rather than scoring candidates independently. Built the evaluation pipeline over web-retrieved evidence, with open-weight models served using vLLM.
- JCA was the most accurate method in all 9 combinations tested (3 instruction-tuned open-weight models × 3 evaluation settings), by 2–11 points over the best baseline, and lost less accuracy than any standard baseline when retrieval favored a wrong answer.

Robustness of RAG systems to the number of retrieved passages ICML 2025

- Showed that evaluating RAG at a single retrieval depth hides brittleness: as more passages are added, some models peak early and then degrade, while others improve and then plateau.
- Introduced metrics for stability (how steady performance stays near the best depth) and scalability (how much a model gains from more passages). Models more easily distracted by irrelevant passages tended to peak early and degrade, a pattern that held across retrievers, rerankers, and prompts.
- Led the experimental design and analysis, and built the retrieval and reranking stack (BM25, dense retrievers with FAISS, cross-encoder rerankers).

Gaming explanation-faithfulness benchmarks Findings of EACL 2024

- Showed that explanation-faithfulness benchmarks can be gamed to produce high scores without making explanations any more faithful.
- On ERASER, a simple wrapper that detects the benchmark's masked inputs inflated scores sharply without changing predictions or explanations on real inputs. On EVAL-X, encoding the predicted label in the explanation achieved the maximum score, though the benchmark was designed to block this shortcut.

INDUSTRY EXPERIENCE

Meta | Research Scientist Intern, Ads Ranking May–August 2025

- Developed a compact representation of long user-behavior histories for ads ranking: older history is compressed while recent events are kept in full, to fit memory and serving-throughput limits.
- Trained ranking models with this representation; in offline evaluation on recent data, it improved normalized entropy (the standard offline ads-ranking metric) by 0.03% relative, above the internal threshold for a meaningful gain, at under 2% throughput cost.

PUBLICATIONS

** equal contribution*

When the Rule Is Clear but the Case Isn't: How User Claims Shape LLM Advice
Jennifer Hsia, Zachary C. Lipton, Aarti Singh. *In submission*

Joint Constraint Aggregation for Qualifier-Sensitive Retrieval-Augmented QA
Jennifer Hsia, Zachary C. Lipton, Aarti Singh. *EMNLP 2026* · Paper · Code

RAGGED: Towards Informed Design of Scalable and Stable RAG Systems
Jennifer Hsia*, Afreen Shaikh*, Zhiruo Wang, Graham Neubig. *ICML 2025* · *Earlier version: oral, NeurIPS 2024 Workshop on Adaptive Foundation Models* · Paper · Code

Goodhart's Law Applies to NLP's Explanation Benchmarks
Jennifer Hsia, Danish Pruthi, Aarti Singh, Zachary C. Lipton. *Findings of EACL 2024* · Paper · Code

An Experimental Study of Balance in Matrix Factorization
Jennifer Hsia, Peter Ramadge. *CISS 2021* · Paper

TECHNICAL SKILLS

Evaluation: LLM-as-judge design and validation against human labels; human annotation and inter-annotator agreement; statistical testing (bootstrap confidence intervals, paired tests)

Tools: Python, Hugging Face Transformers, vLLM, multi-GPU inference on Slurm, frontier-model APIs (OpenAI, Anthropic, Gemini), retrieval (BM25, dense retrievers, FAISS)

TEACHING & SERVICE

Teaching Assistant, CMU: Human-AI Complementarity for Decision Making (2025); Probabilistic Graphical Models (2023)

Machine Learning Department TA Award (one of four TAs recognized for 2021–2023) 2023

Workshop Organizer: Human Evaluation of Generative Models, NeurIPS 2022

Reviewer: ACL, EMNLP, ICML